

Hidden Challenges in Evaluating Spillover Risk of Zoonotic Viruses using Machine Learning Models

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Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This is an interesting manuscript demonstrating an innovative re-purposing of pre-trained large language models for zoonotic risk prediction. That, along with comparisons to a range of existing approaches are major strengths. However, there are issues with the training data being used which means conclusions around model performance and generalisability are not supported at present (pseudo-replication, label definition, and dataset curation, see below). In discussing the challenges of zoonotic risk prediction, the manuscript also suffers from a frequent use of anecdotes over systematic comparisons, particularly selecting a few pandemic viruses which are miss-predicted when both previously-published results and current performance metrics suggest most viruses are accurately predicted. Since none of these models claim to predict pandemic potential or virulence (merely ability to infect), this seems like an unfair criticism.

Pseudo-replication:

Line 373 - 378: From this description it seems the data are pseudo-replicated, with potentially large numbers of nearly-identical sequences spread across training and test data. Simply dividing the data based on collection date as is done for some families in later tests is also not a robust test of generalisation in this situation - if in future we discover a sequence very similar to a known human-infecting virus, we can already conclude that it may infect humans if given the opportunity. The challenge is to predict truly novel viruses (e.g. new species or genera), so the holdout dataset should be phylogenetically-distinct, not just temporally-distinct. This issue has been raised repeatedly by other authors, and indeed is a common criticism of data-hungry deep learning based approaches which should be addressed in some way in this manuscript. See e.g. Mollentze & Streicker, 2023 (<https://doi.org/10.1016/j.coviro.2023.101346>), Brierley & Fowler, 2021 (<https://doi.org/10.1371/journal.ppat.1009149>), Mollentze et al, 2021 (<https://doi.org/10.1371/journal.pbio.3001390>), Young et al, 2020 (<https://doi.org/10.1371/journal.pcbi.1007894>).

Line 155 - 160: This is where the potential pseudo-replication becomes an issue - is the improved performance of existing approaches due to a larger dataset, or simply due to closely-related genomes being in both the test and training data? At least some of the published methods being compared specifically aimed to avoid this, to give accuracy measures which reflect real-world performance on novel virus species, but these will necessarily be much lower than what can be achieved here.

Line 228 - 229: As above, good model performance in itself is not sufficient to claim that the model is generalisable. For that, we would need to know how distinct the test data are from the training data.

Label definition:

Line 123 - 126: From this description, it seems that the infectivity label actually reflects the sampled host from which the virus sequence was generated, not zoonotic ability more broadly. By definition zoonotic viruses will be found in more than one host species, so how is this resolved?

Dataset curation:

Line 128 - 129, 181 - 183, 364 - 367: Please describe what (if any) curation was done. At present, it sounds like the authors simply downloaded the NCBI virus database, which is already curated by others. If that is the case, then it cannot be claimed as a novel data source.

Line 369 – The way segmented viruses are handled means the data contains artificial re-assortant viruses which do not

occur in nature. No conclusions about performance on segmented viruses are safe when this is the case. There are also several other issues here: (1) it should not be possible for an isolate to have more sequences than segments – is “isolate” the correct term?; (2) why sample two sequences per segment when non-segmented viruses are represented by a single genome, and what happens when one copy ends up in the test data while the other is in the training data?; (3) there are other segmented virus families in the data beyond the 12/8/3 segment ones listed, e.g. Arenaviridae (2 segments), Sedoreoviridae (10 - 12 segments depending on species).

Other comments:

Line 45 - 49, 193 - 195: Given how the model was evaluated, I think there should be a caveat that we currently only know that the model works on reads/contigs already identified as viral. How the model will behave with completely unknown contigs which may represent a mixture of viruses and other organisms remains unknown.

Line 83 - 86: That unsupervised feature extraction is needed has not been established. Please add further arguments and/or citations on this point. In particular, how do we explain the good performance of existing approaches which involve manual feature engineering (including the performance demonstrated for such methods in this manuscript)? Further, the cited publication uses manual feature engineering, not large language models – is this the intended citation?

Line 104 - 106: It is unclear to me what the issue identified here is. As written, this could mean either (a) the model is wrong on some lineages (but this is expected, since no model will be perfect), or (b) predictions are too coarse-grained to distinguish zoonotic from non-zoonotic lineages (and then I would question whether this is really a problem - e.g. we know SARS virus as a species is high risk, and have little evidence that other SARS lineages won't also emerge in future).

Line 147 - 148: The support for the hypothesis of immune evasion through genome mimicry of host genomes is overstated here. Beyond the optimisation of CpG to evade Zinc-finger antiviral protein (ZAP), we do not yet know whether such mechanisms are common, nor do we know of species-specific examples that would lead to selection for genomic signals specific to individual host species. Some of the citations used here appear to be pointing to the wrong references.

Line 165 - 167, 424 - 427: The way the BLAST-based baseline is set up seems unnecessarily strict and unrealistic, creating a baseline which is weaker than it should be. We know that there will be gaps in surveillance, and the machine learning models can generalise across much broader phylogenetic patterns (e.g., all sequences from the same virus species). I would suggest a baseline that does the same, e.g. predicting a virus as human-infecting if any of its close BLAST-hits were sampled from humans. Further, removing clearly zoonotic viruses (the same or a very similar genome found in both humans and other animals, as indicated by identical bitscores) as “unpredictable” leaves only the more difficult cases for the BLAST-based baseline, when other models are not subject to this restriction.

Line 243 - 246: How systematic is this failure to identify zoonotic lineages? If model performance is good, then it can't be that bad. Some mistakes are expected, so please show evidence that errors are taxonomically or phylogenetically-biased to support the claim that entire lineages are unpredictable. Some other questions which should be addressed: What is the cause? How well-represented are these lineages (or close relatives) in the training data? Do we just need more virus discovery and/or labelled data? That would be a problem that cannot be solved with better models.

Line 257 - 259: As above, a more systematic comparison would be needed to support such a broad conclusion. At present, it is unclear why these particular lineages were evaluated, and it is essentially a sample size of 2. For a more systematic comparison, one option might be to compare model accuracy to a quantitative measure of how frequently each family switches its ability to infect humans (perhaps something similar to the measure used by Geoghegan et al, 2017 - <https://doi.org/10.1371/journal.ppat.1006215>).

Line 274: The results focus almost entirely on sensitivity, i.e. accurate prediction of known human-infecting viruses. But what about specificity (exclusion of viruses which do not infect humans)?

Line 323 - 327: The claim that current models cannot predict “most zoonotic viruses as high risk” seems hard to believe. Since most human-infecting viruses are zoonotic, how are the very high model performance metrics achieved if this statement is true? Please add a systematic comparison to support this claim.

Line 358 - 360: What are the valuable perspectives on enhancing generalization capabilities that were identified? Without identifying the causes of a lack of generalization (see comments for line 243 - 246), how can we resolve them?

Line 441 - 443: Please explain how and why this downsampling was done. As written, it sounds like poorly-predicted families are being removed, but what test data are being used to evaluate this performance? Are subsequent performance metrics still reliable when this is done?

Reviewer #2

(Remarks to the Author)

The authors present a thoroughly constructed study, particularly in its innovative use of large language models, and the systematic validation protocols.

Main issues:

Labelling of data: As the authors state: "Because the human infectivity of many viruses has not been directly validated, the infectivity label (i.e., human-infecting or animal-infecting) for each viral strain was defined based on the host information from which the viral sequences were collected", have the authors verified (either all or subsample) their virus-host data to identify whether the associations are more likely to be virus infects/affects host, rather than virus associates with host. For instance, a virus may have had e.g. humans labelled as the host, but the sample could have been from stomach content? Have the authors accounted for this in their data?

Generalisability: I am not sure if the hypothesis that "a queried virus would have the same infectivity as the most similar viral sequence" or if that hypothesis works particularly well with emerging viruses.

As the authors mention, generalizability is correlated with virus sequence similarity. However, I am not certain curating viral lineages associated with zoonoses would enhance generalizability. How do the authors envision this to work? Particularly given the fact that our knowledge is extremely biased: we are more likely to study any virus species/genus/family if it contains a human-infecting/zoonotic virus than if it does not. And this knowledge is often constructed post an emergence event.

Minor points:

For labelling of animal-infecting or human-infecting, this is applied at the sequence level, I suggest the authors revise the manuscript to clarify, as the same strain of a virus could be both animal and human infecting.

The authors should note that there exist data sources that have harvested taxonomically mapped virus-host (pathogen-host) associations from GenBank.

Reviewer #3

(Remarks to the Author)

Summary

In this work, the authors attempt to build improved machine learning models to predict virus human infectivity potential by creating a substantial, new curated dataset including 26 families and fine-tuning BERT LLMs pre-trained either on the human whole genome or on NCBI RefSeq viral database. They perform thorough benchmarking, comparing their models with several other models with distinct architectures (including kNN, CNN and GBM) that they re-train on their dataset. They find that performances generally improve across all models following re-training using their new curated dataset showing the importance of the size, quality and comprehensiveness of the training dataset for prediction. In their comparison, the authors find that their fine-tuned models, BERT-infect DNABERT and BERT-infect ViBE, show the best performances compared to others on the "past" virus dataset, when applied to high-throughput sequencing data, and on the "future" virus dataset. However, when considering the enrichment in positive labels in the top 20% of predicted probabilities - a metric used to evaluate the capacity for prioritization of high-risk viruses - all models appear to perform equally. Finally, the authors identify that all models perform poorly for specific viral families e.g. (Rhabdoviridae, and Hantaviridae) and genera (Betacoronavirus, Flavivirus), highlighting severe limitations in terms of human infectivity prediction capacity.

General Comment

This study provides valuable improvements for the development of ML models for the prediction of virus human infectivity (i) by providing a comprehensive, curated dataset that appears to generally improve model performance across several types of architecture (ii) through their benchmarking work, comparing their models to at least three other ones, (iii) by highlighting important limitations for human infectivity prediction overall. At this stage I have several comments:

Major Comments

Definition of host infectivity in the curated dataset

In the curated dataset used for model fine-tuning, each sequence is associated to a specific host, which may lead to important biases in the input data. For most virus species, sequencing data is strongly biased as sampling is uneven across geographic origins and host species and as some hosts are more easy to sample/access than others and as sequencing capacities may vary across locations. This will likely result in a training dataset that does not reflect the true ecology of the virus, in particular for viruses with a complex ecological cycle including arboviruses and may explain the results obtained for e.g. the Flavivirus genus. How do the authors balance the representation of hosts for viruses with two (or more) host species?

Unidentified human-infecting viruses and false positives

The authors mention that they do not use experimentally validated data which results in two major limitations (i) input data labeled as not human for viruses with undemonstrated human-infection capacity ; (ii) a difficulty to discriminate between false and real positives. To better evaluate the validity of the authors' strategy (of using data not validated experimentally), could the authors try to evaluate the impact of having 5 or 10% additional unlabelled human-infecting viruses by artificially removing those labels for some known viruses ?

Use of detailed viral phylogenies

The authors use viral phylogenies to compare F1 scores and host labels for the Flaviviridae and Coronaviridae families but the trees are difficult to read. For a genus as complex in terms of ecology as the Flavivirus genus, it would be beneficial to provide a more detailed view of the phylogeny to better identify the F1 score distribution along the tree.

Minor comments

Use of VIDHOP

It is unclear to me why the authors drop the VIDHOP model from their analysis.

228-229

“These results demonstrate a high level of model generalization toward novel viruses”

Please, provide references to support your evaluation that the level of model generalization is indeed high.

333-335 “Models using nucleotide sequences as inputs may effectively detect host adaptations in viral genome compositions; however, capturing infectivity changes due to minor genomic variations remains challenging.”

The term “minor” is extremely context-specific, a single change can have a dramatic effect on a protein structure and the capacity to replicate within a given host. I recommend the authors reformulate towards a more objective term e.g. “a small number of genetic variations”.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have addressed my previous concerns, and the manuscript is much improved. I only have relatively minor comments remaining:

The new version introduces the phrase “newly emerged viruses” when referring to the future virus dataset. I think this overstates what this dataset represents, and by extension what the models evaluated are able to predict. All we know is that the specific genotypes are newly discovered / recently sequenced, there is no indication that these genotypes represent emerging viruses. For example, if someone sequenced and published a new rabies virus genome in 2020, then it would end up in the future virus test set. However, we already know about this virus more broadly (and it is probably represented in the training data), so the new genome does not make it a new virus, and certainly does not qualify it as an emerging virus.

The finding that current models do not predict zoonotic H5Nx influenza viruses (Discussion, line 324 – 325) should be mentioned in the results too, as it supports the point about SARS-CoV-2 also being missed by current models.

Other comments:

Line 129 – 132: Should this be <20 ? Otherwise the wording here is unclear.

Line 313 – 316. I think this cross-reference should be to supplementary figure S4.

Line 437: I think this should read $<1e-4$ (not $>1e-4$).

Line 532 – 533: Typo in “real-world”.

Supplementary figure 1: From the caption, it is unclear whether points represent individual viruses (i.e., one per virus family), or a summary of predictions across multiple viruses in each family.

Reviewer #2

(Remarks to the Author)

The authors have thoroughly responded to most of my initial comments, and I believe the manuscript is much better overall after the first round of reviews.

The only point that is still bit ambiguous to me is in relation to labelling of data. Perhaps I was not so clear in my previous attempt, but I did not only mean environmental samples (those that are either without hosts, or labelled as environmental samples in GenBank), but also extends to samples with a well-defined, non-ambiguous host, e.g. in GenBank there exists few instances of plant viruses, and other non-human infecting viruses, that are associated with humans (e.g. host species indicated as human) due to them being sequenced from stomach content, or skin samples, etc. Have the authors considered this issue (in addition to removing environmental samples?). In such cases, the host will not be ambiguous, but the association with humans (or the host species) is simply commensal.

Reviewer #3

(Remarks to the Author)

Main comment

The authors addressed the main concerns raised by the reviewers and revised their paper substantially, resulting in a paper that is now easier and clearer to read, with a better delineation of the achievements and perspectives of the study.

Specifically, the authors revised their paper by removing their claims regarding the generalisation capacity of their model in particular discussing its limits regarding the evaluation of infectivity for novel viruses. They also more thoroughly discuss the limits of their labeling approach and the need for improving labeling more generally to better handle the question of dual/multi-host viruses. Also, they re-evaluated their baseline based on the BLAST model to make it more realistic, and better detail their methods regarding the handling of segmented viruses and the curation of Genbank data. I still have a few comments that would need to be addressed for the manuscript to be suitable for publication.

Specific comments:

As discussed in the rebuttal in the response to several of the reviewer's comments, one of the main aims of this study is the improvement of sensitivity (ability to detect virus with human infecting potential) rather than specificity I think briefly clarifying this point in the text would be beneficial to better depict the aims of this study.

As mentioned in one of the authors' responses, combining infectivity prediction to other approaches including surveillance in animals and experimentation will remain essential to preparedness against zoonotic threats. I suggest adding a few words on this question in the discussion to provide a broader perspective on the general integration of such approaches in a more tangible public health perspective.

I320-322: "Furthermore, another limitation in our dataset curation is that the data labeling method cannot distinguish between infectivity (i.e., capability for animal-human infection) and transmissibility (i.e., ability to expand human-human infections)"

Of note, for animals feeding on other animals/humans, detection in a given organism may not even represent infection capacity (the virus can be detected in a blood meal of e.g. a mosquito that fed on an infected host but is resistant to infection), please could you reformulate the sentence accordingly ?

Version 2:

Reviewer comments:

Reviewer #2

(Remarks to the Author)

The authors have further improved their manuscript, and I have no further substantial comments or suggestions to add. The work itself is solid and addresses an important gap in research.

Reviewer #3

(Remarks to the Author)

The authors have responded to all my comments, further improving their manuscript. I have no remaining comments to make.

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Reviewer Comments:

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Response: First, we appreciate the reviewer's positive impression of our manuscript and the insightful and meaningful comments, which have helped us improve substantially improve the manuscript.

This comment raises concerns about the generalizability and biased analyses of zoonotic viruses. First, as the reviewer pointed out, describing our results using the term "generalization" was inappropriate, and taxonomic hold-out validation should be performed to make claims about generalization. Therefore, we have removed the word "generalization" from the manuscript and revised the content to avoid misleading statements: similar to previous research, this study was focused on constructing a model using available virus sequences at a given time point to then predict the infectivity of viruses that emerged afterward. To clarify this focus, we have modified the following sentences: lines 49-50, 63-65, 98-100, 137-140, 199-202, 220-221, 252-254, 313-317, 389-391, and 546.

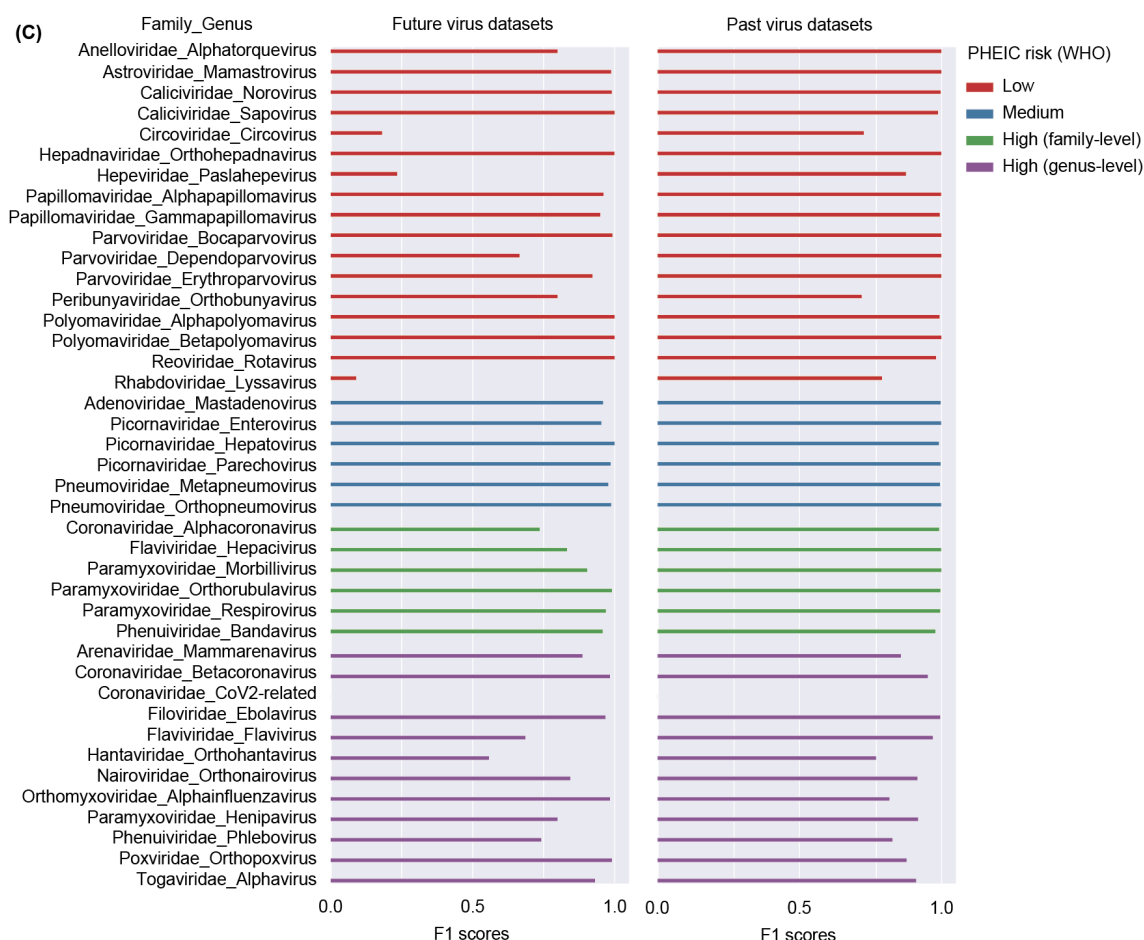
Next, we would like to emphasize that identifying the difficult-to-predict viral lineages did not result from biased analyses. In Supplementary Fig. 4, we highlighted that while our study could accurately predict the infectivity of most future viral data, prediction for some virus genera was

challenging. As described in the figure legend, we listed difficult-to-predict viral genera whose F1 scores were below 0.75 in the future viral datasets, which were not exclusively focused on pandemic-causing viruses (see **lines 223-241**). In other words, the key issue revealed by our study is that the list of difficult-to-predict viral lineages based on F1 scores contained important pandemic-related viral lineages.

Moreover, we have provided additional validation in the revised manuscript: a comparison between our results shown in Supplementary Figs 4A-B and the pandemic risks evaluated by the WHO (<https://www.who.int/publications/m/item/pathogens-prioritization-a-scientific-framework-for-epidemic-and-pandemic-research-preparedness>). It is noteworthy that half of the difficult-to-predict viral genera (those with an F1 score <0.75 in the future virus datasets) were part of high-risk families listed by the WHO (i.e., Coronaviridae, Flaviviridae, Phenuiviridae, and Hantaviridae). We believe that these findings can lead to future pandemic preparedness through predictive model improvements. We have therefore added this figure as Supplementary Figure 4C and revised the manuscript as follows:

Lines 233-238: Notably, half of these difficult-to-predict viral lineages were listed as high-risk viruses by the WHO pathogen prioritization (Supplementary Fig. 4C).

Lines 606-608: c WHO risk assessment for difficult-to-predict viral lineages. The colors are based on the WHO PHEIC (Public Health Emergency of International Concern) risk levels. Viruses listed as priority pathogens are represented in green and purple, based on the genus and family to which they belong.



Pseudo-replication:

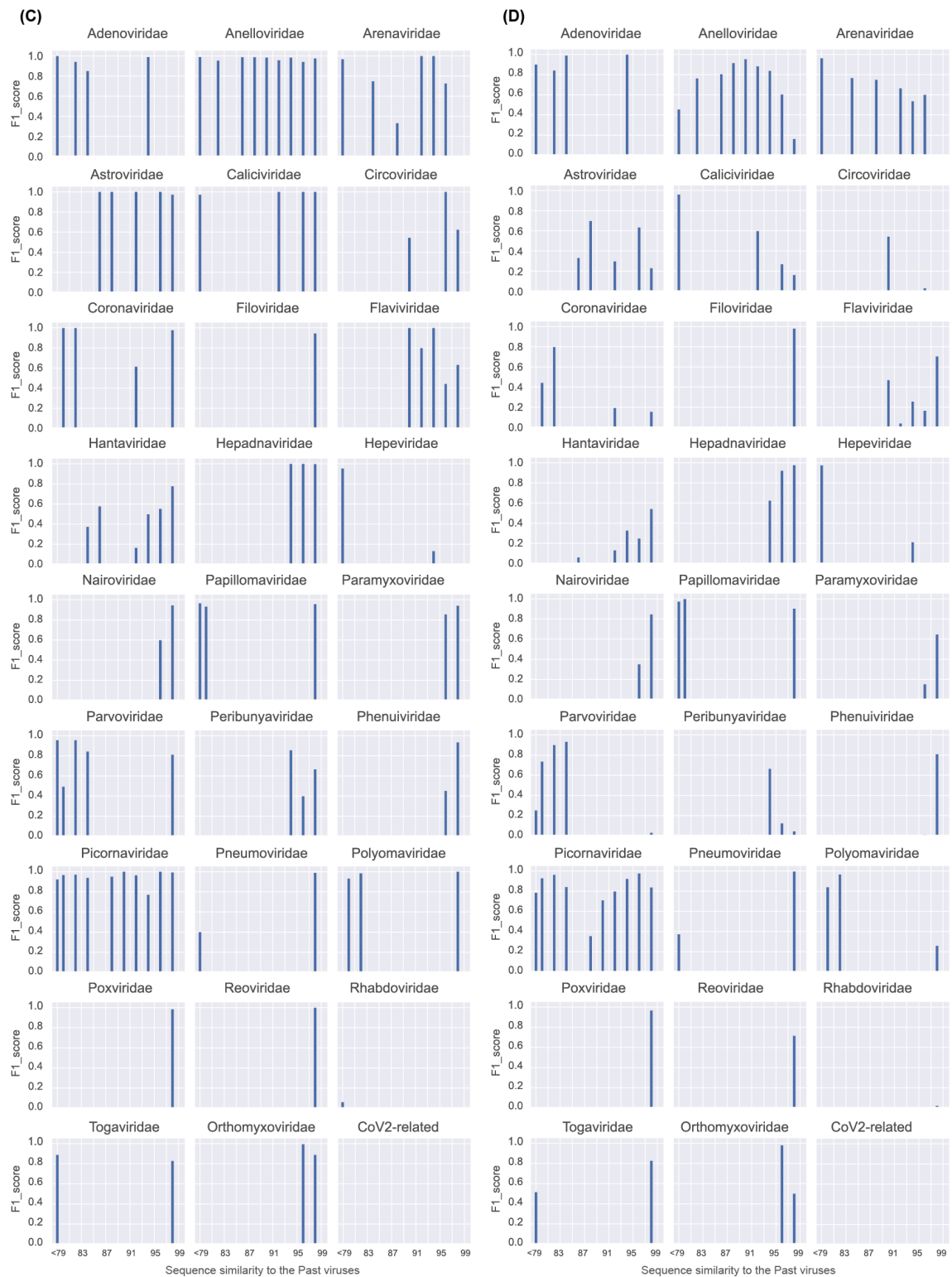
Line 373 - 378: From this description it seems the data are pseudo-replicated, with potentially large numbers of nearly-identical sequences spread across training and test data. Simply dividing the data based on collection date as is done for some families in later tests is also not a robust test of generalisation in this situation - if in future we discover a sequence very similar to a known human-infecting virus, we can already conclude that it may infect humans if given the opportunity. The challenge is to predict truly novel viruses (e.g. new species or genera), so the holdout dataset should be phylogenetically-distinct, not just temporally-distinct. This issue has been raised repeatedly by other authors, and indeed is a common criticism of data-hungry deep learning based approaches which should be addressed in some way in this manuscript. See e.g. Mollentze & Streicker, 2023 (<https://doi.org/10.1016/j.coviro.2023.101346>), Brierley & Fowler, 2021 (<https://doi.org/10.1371/journal.ppat.1009149>), Mollentze et al, 2021

(<https://doi.org/10.1371/journal.pbio.3001390>), Young et al, 2020
(<https://doi.org/10.1371/journal.pcbi.1007894>).

Response: We agree with the reviewer's comment that pseudo-replication datasets could lead to overestimating model performance. To check for this issue, we investigated model performance for each sequence similarity between training and test datasets and confirmed that the BERT-infect_{DNABERT} model outperformed the BLAST baseline model even on data with low similarity between the train and test sets, although model performances tended to be higher when the sequence similarity between training and test data was high (details are provided later: see comment on lines 228-229). We have added this result as Supplementary Figure 2C-D and modified the manuscript as follows:

Lines 593-595: Comparison of prediction performance based on sequence similarity between training and test data sets: the F1 scores of the BERT-infect_{DNABERT} and BLAST model are plotted in **c** and **d**.

Furthermore, as pointed out by the reviewer, hold-out validation at the genus or species levels should be performed to describe model generalization capability. To clearly indicate the focus of our study, we have removed the term "generalization" from the manuscript.



Line 155 - 160: This is where the potential pseudo-replication becomes an issue - is the improved performance of existing approaches due to a larger dataset, or simply due to closely-related

genomes being in both the test and training data? At least some of the published methods being compared specifically aimed to avoid this, to give accuracy measures which reflect real-world performance on novel virus species, but these will necessarily be much lower than what can be achieved here.

Response: We appreciate the constructive comment, according to which we have reconsidered the comparison in Supplementary Figure 1 and deemed it inappropriate because we compared models trained on different datasets; it is thus unclear whether improved performance was due to using a new dataset or simply due to the difference in dataset predictability. Therefore, we have removed Supplementary Figure 1 and the relevant claims from the manuscript.

Line 228 - 229: As above, good model performance in itself is not sufficient to claim that the model is generalisable. For that, we would need to know how distinct the test data are from the training data.

Response: We apologize for the repetition. We have reconsidered use of the term "generalization" to describe the focus of this study and deemed it inappropriate; we have therefore removed this term from the manuscript.

We also apologize for repeating the response to the comments on lines 373-378. We agree with the reviewer's comment that it is important to verify how model performance changes based on similarity between the training and test datasets, as this also relates to the issue of pseudo-replication within the dataset. Thus, we additionally validated the relationship between sequence similarity and prediction performance between the training dataset (past viruses) and the test dataset (future viruses). We have added this result as Supplementary Figure 2C-D.

Label definition:

Line 123 - 126: From this description, it seems that the infectivity label actually reflects the sampled host from which the virus sequence was generated, not zoonotic ability more broadly. By definition zoonotic viruses will be found in more than one host species, so how is this resolved?

Response: We appreciate this reviewer's constructive comment and agree that there is room of improving the labeling method. In this study, the labeling method was designed to detect the human infectivity of zoonotic viruses, following the same strategy as that employed in previous studies; however, we also believe that using multi-labeling could improve model performance. On the other hand, taxonomic hierarchy (species, subspecies, etc.) used for multi-labeling is a critical hyperparameter, and this issue has not been investigated, as pointed out by Mollenze et al. in their review paper. Thus, as a future direction, it is necessary to conduct preliminary investigations into how much sequence variation leads to infectivity changes and then set the appropriate hierarchy for building predictive models. We have added the following statement to the Discussion section.

Lines 302-308: Furthermore, the multi-labeling method, which explicitly indicates viruses that infect both animals and humans, may improve predictive performance for zoonotic viruses. On the other hand, an important hyperparameter for the multi-labeling method has not been investigated: the taxonomic hierarchy to assign labels. Future work will need to include preliminary investigations into the degree of sequence variation that leads to infectivity changes to identify the appropriate taxonomic hierarchy for multi-labeling.

Dataset curation: Line 128 - 129, 181 - 183, 364 - 367: Please describe what (if any) curation was done. At present, it sounds like the authors simply downloaded the NCBI virus database, which is already curated by others. If that is the case, then it cannot be claimed as a novel data source.

Response: We appreciate the reviewer for pointing insufficient explanations. In this study, we curated the datasets for the viral infectivity prediction model through several steps. First, to eliminate redundancy within the dataset, we removed sequences that were 100% identical and had the same infectivity label for non-segmented viruses. For segmented viruses, we grouped sequences into viral isolates based on the combination of 22 entries in the metadata and then manually checked their genome set (following the reviewer's comment on line 369, we have added information on the manual curation step; please see details below). While the downloaded

data from the NCBI Virus database contained 1,336,901 sequences, the curated data contained 140,638 sequences. We have added the following information to the Methods section:

Lines 384-385: Through these dataset curations, we selected 140,638 sequences from the 1,336,901 sequences downloaded from the NCBI Virus Database.

Line 369 – The way segmented viruses are handled means the data contains artificial re-assortant viruses which do not occur in nature. No conclusions about performance on segmented viruses are safe when this is the case. There are also several other issues here: (1) it should not be possible for an isolate to have more sequences than segments – is “isolate” the correct term?; (2) why sample two sequences per segment when non-segmented viruses are represented by a single genome, and what happens when one copy ends up in the test data while the other is in the training data?; (3) there are other segmented virus families in the data beyond the 12/8/3 segment ones listed, e.g. Arenaviridae (2 segments), Sedoreoviridae (10 - 12 segments depending on species).

Response: We appreciate the reviewer’s constructive comment. We have added information about the manual curation step for preparing segmented viral datasets. While the comparative model performance results did not change significantly, we were able to correct mistakes in dataset construction (relevant for Figs. 1C, 2C-D, and 3B-C).

In this study, we clustered virus sequences based on common metadata to link the genome sets of segmented viral isolates. This assumed that "sequences registered by the same authors at the same time and obtained from the same host organism are likely to originate from the same viral isolate." However, as pointed out in (1), there was a possibility of unnatural reassortant viruses, so we added a manual curation step: we checked the segment information in the metadata to ensure that multiple segment sequences were not assigned to a single viral isolate. Through this curation, we removed 76 virus isolates.

Additionally, as the comment (2), the number of genome segments is known to vary widely across viral families. In this study, we checked the segment column in the metadata and excluded viruses with more genome segments than the reported maximum for each viral family. This curation method allowed for inclusion of viruses such as arenaviruses with two segments and sedoreoviruses with 10 segments.

Regarding (3), our datasets were split at the virus isolate level, ensuring that no sequences from the same virus isolate were included in both the training and test datasets. This prevented data leakage, which the reviewer was concerned about (the assignment of each virus isolate to the training, validation, and test datasets in the five-fold cross-validation is described in Supplementary Table 1).

According to the reviewers' constructive comment, we revised the Methods section as follows:

Line 377-382: If a single viral isolate contained more sequences than the specified number of viral segments (i.e., Orthomyxoviridae: 8, Rotaviridae: 12, and other segmented RNA viruses: 3), redundancy was eliminated by randomly sampling a sequence for each segment. We also checked the Segment column in metadata to ensure that multiple segment sequences were not assigned to a single virus, which resulted in the removal of 76 virus isolates and 1,270 sequences.

Other comments:

Line 45 - 49, 193 - 195: Given how the model was evaluated, I think there should be a caveat that we currently only know that the model works on reads/contigs already identified as viral. How the model will behave with completely unknown contigs which may represent a mixture of viruses and other organisms remains unknown.

Response: As pointed out in the reviewer's comment, we assumed that the input for the prediction model was a sequence that had already been identified as a virus. We have added this clarification as follows:

Lines 45-49: Our model also exhibited high predictive performance even with partial viral sequences, such as high-throughput sequencing reads or contig sequences from *de novo* sequence assemblies, indicating the model's applicability for mining zoonotic viruses from virus metagenomic data.

Lines 185-188: These results indicated that only certain models are suitable for mining human-infecting viruses when inputting partial viral sequences (i.e., high-throughput sequencing reads or assembled contigs).

Line 83 - 86: That unsupervised feature extraction is needed has not been established. Please add further arguments and/or citations on this point. In particular, how do we explain the good performance of existing approaches which involve manual feature engineering (including the performance demonstrated for such methods in this manuscript)? Further, the cited publication uses manual feature engineering, not large language models – is this the intended citation?

Response: As pointed out in the reviewer's comment, overemphasized the need for unsupervised feature extraction. We have thus revised the term to "helpful." Additionally, citation 12 was mistakenly cited instead of 11, so this has been corrected.

Lines 83-85: Unsupervised feature extraction from viral genetic sequences and their interpretation are helpful in understanding the mechanisms driving viral infectivity as well as improving model performance.

Line 104 - 106: It is unclear to me what the issue identified here is. As written, this could mean either (a) the model is wrong on some lineages (but this is expected, since no model will be perfect), or (b) predictions are too coarse-grained to distinguish zoonotic from non-zoonotic lineages (and then I would question whether this is really a problem - e.g. we know SARS virus as a species is high risk, and have little evidence that other SARS lineages won't also emerge in future).

Response: As pointed out by the reviewer, it is expected that some viral lineages may be more difficult to predict; however, we highlight an important issue in preparedness for future zoonotic viral diseases in that high-pandemic-risk virus lineages defined by the WHO were included among the difficult-to-predict-lineages (relevant to the first comment of reviewer 1). We believe that pointing out these unresolved issues will lead to future model improvements.

Line 147 - 148: The support for the hypothesis of immune evasion through genome mimicry of host genomes is overstated here. Beyond the optimisation of CpG to evade Zinc-finger antiviral protein (ZAP), we do not yet know whether such mechanisms are common, nor do we know of

species-specific examples that would lead to selection for genomic signals specific to individual host species. Some of the citations used here appear to be pointing to the wrong references.

Response: We appreciate this reviewer's detailed check. Following the comment, we have revised the manuscript to tone down statements and corrected the citation number.

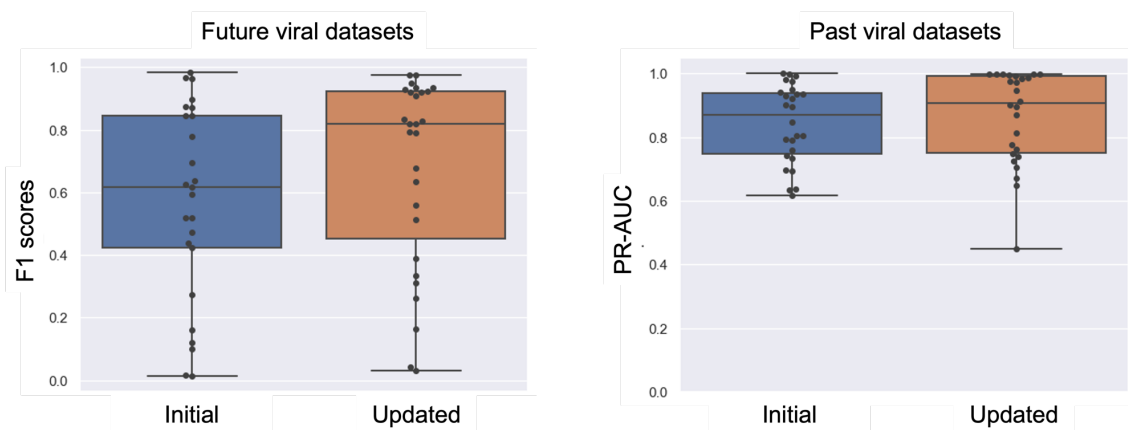
Lines 146-149: Molecular mimicry of host organisms is known as a factor that defines host range by contributing to efficient replication and immune evasion of viruses, and therefore, we hypothesized that the DNABERT model, pre-trained with the human genome, could extract representative features associated with human infectivity.

Line 165 - 167, 424 - 427: The way the BLAST-based baseline is set up seems unnecessarily strict and unrealistic, creating a baseline which is weaker than it should be. We know that there will be gaps in surveillance, and the machine learning models can generalise across much broader phylogenetic patterns (e.g., all sequences from the same virus species). I would suggest a baseline that does the same, e.g. predicting a virus as human-infecting if any of its close BLAST-hits were sampled from humans. Further, removing clearly zoonotic viruses (the same or a very similar genome found in both humans and other animals, as indicated by identical bitscores) as “unpredictable” leaves only the more difficult cases for the BLAST-based baseline, when other models are not subject to this restriction.

Response: We appreciate the insightful comment that has helped to substantially improve the manuscript. As pointed out in the reviewer's comments, the initial BLAST model could have been a weak baseline model, so we performed additional validation for a new BLAST baseline model. In the new BLAST model, if a human-infecting virus was included among the hit sequences, it was predicted to be human-infecting. We thus found that the prediction performance of the new BLAST model improved for both past and future viral datasets. However, it is important to note that the updated BLAST baseline model did not outperform machine learning models, including the BERT-infect models. We have revised the manuscript accordingly:

Lines 157-159: BLASTn was used to assess the potential predictability of our datasets based on the simple hypothesis: if a human-infecting virus was included among the hit sequences, it was predicted to be human-infecting (see Methods).

Lines 433-438: We used BLASTn to evaluate potential dataset predictability based on the simple hypothesis that a virus showing sequence similarity to human infectious viruses is predicted to be human infectious. BLASTn prediction was conducted using the training dataset as a database and test dataset as a query. The following cases were judged as unpredictable: (i) there was no hit sequence with an E-value of $>1e-4$, or (ii) the aligned length of the top hit sequence did not cover $>50\%$ of the query sequence.



Line 243 - 246: How systematic is this failure to identify zoonotic lineages? If model performance is good, then it can't be that bad. Some mistakes are expected, so please show evidence that errors are taxonomically or phylogenetically-biased to support the claim that entire lineages are unpredictable. Some other questions which should be addressed: What is the cause? How well-represented are these lineages (or close relatives) in the training data? Do we just need more virus discovery and/or labelled data? That would be a problem that cannot be solved with better models.

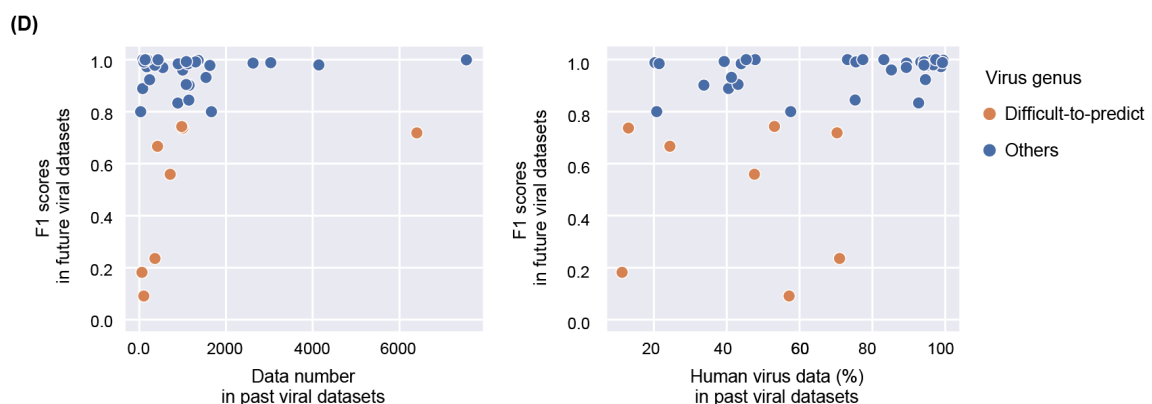
Response: We appreciate this comment, which highlights two major points: (1) phylogenetically systematic evaluation for difficult-to-predict viral lineages and (2) causes of difficult-to-predict viruses.

Regarding the first point, Supplementary Figure 4 compares the difficulty of predicting zoonotic virus lineages at the genus level. These results are explained in **lines 223-241**. Even if the family-level prediction performance was apparently high, there was a tendency for lower performance in predicting specific viral genera, as shown in Supplementary Figure 4A.

Regarding the second point, we fully agree with the reviewer's comment that pursuing the cause for prediction difficulty is important for improving the model; however, we could not identify clear reasons at this time. In the present, we compared the predictive performance at the genus level with (1) the data volume or (2) the proportion of human-infecting viruses in the training datasets, respectively. For both indicators, no significant differences were observed among viral lineages.

A potential explanation could be that when small amino acid changes lead to changes in infectivity, it can be difficult for genome-based machine learning models to make accurate predictions. We are currently working on predicting viral infectivity using protein language models, and we have observed improved performance for virus lineages that are difficult to predict (subgenus Sarbecovirus and Merbecovirus; data not shown). Recent studies have also reported that protein language models successfully predicted the infectivity of SARS-CoV-2, suggesting that adopting different model architectures could lead to performance improvements (<https://doi.org/10.1101/2023.04.07.536023>).

This result has been added as Supplementary Figure 4d; the following sentences have been added as well.



Line 235-238: Although we checked for problems in the training dataset for these difficult-to-predict virus lineages, neither the amount of data were extremely small, nor the proportion of viruses that infect humans was particularly low (Supplementary Fig. 4D), which means that the reason behind the difficulty of prediction could not be clarified (see Discussion).

Line 334-341: We identified difficult-to-predict viral lineages, but we found no problems with these viral genera in the training datasets; the reason behind this difficulty of prediction thus remains unclear (**Supplementary Fig. 4D**). It may be that frequent changes in virus infectivity during Flavivirus evolution might have hindered model training (**Fig. 4**); however, the association between prediction difficulty and changes in infectivity should be further assessed through virus-host evolutionary analysis. Another possibility is that capturing infectivity changes due to a small number of genomic variations might be challenging for models using nucleotide sequences as the input.

Line 610: d Characteristics of the training dataset for difficult-to-predict viral lineages.

Line 257 - 259: As above, a more systematic comparison would be needed to support such a broad conclusion. At present, it is unclear why these particular lineages were evaluated, and it is essentially a sample size of 2. For a more systematic comparison, one option might be to compare model accuracy to a quantitative measure of how frequently each family switches its ability to infect humans (perhaps something similar to the measure used by Geoghegan et al, 2017 - <https://doi.org/10.1371/journal.ppat.1006215>).

Response: We apologize for the confusion. We revised the sentence to tone down the statement in line with the reviewer's comment.

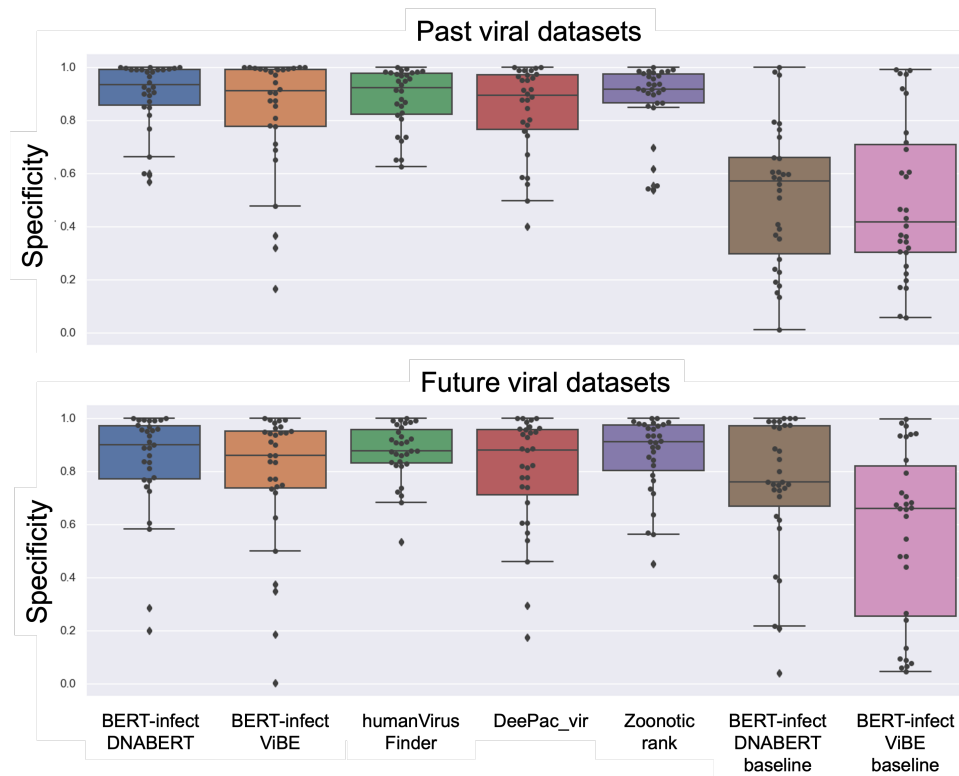
Line 252-254: These results suggest the insufficient predictive capability of current models for viral lineages in which the zoonotic infection potential has been acquired repeatedly.

Additionally, we believe that incorporating viral evolutionary analysis is essential for evaluating the performance of infectivity prediction models, and this is one of the main advancements

provided by this study (as noted in **lines 50-53 and 281-285**). Previous studies have uniformly evaluated model performance for the category "viruses;" however, our validation at the genus level highlights previously overlooked virus lineages that are difficult to predict, contributing to future model improvements. In future work, the quantification of host switching proposed in the referenced paper, as well as analysis of sequence changes related to host switching, as discussed in this paper (<https://www.nature.com/articles/s41559-024-02353-4>), would be useful for improving the model.

Line 274: The results focus almost entirely on sensitivity, i.e. accurate prediction of known human-infecting viruses. But what about specificity (exclusion of viruses which do not infect humans)?

Response: In response to the reviewer's comment, we have presented the results of our specificity calculations. These results show that high specificity was observed across all the models compared. However, as mentioned in **lines 293-308** of the Discussion, animal-infecting viruses were labeled based on "viruses that have not been confirmed to infect humans," so this label does not guarantee that these viruses do not infect humans. Therefore, we primarily focused this study on the ability to detect human infectivity of viruses (i.e., sensitivity).



Line 323 - 327: The claim that current models cannot predict “most zoonotic viruses as high risk” seems hard to believe. Since most human-infecting viruses are zoonotic, how are the very high model performance metrics achieved if this statement is true? Please add a systematic comparison to support this claim.

Response: We appreciate this reviewer for pointing out our misleading explanations. First, it is important to note that most influenza A viruses circulate seasonally in human populations and are not zoonotic. Particularly, H5Nx influenza A viruses have been reported to infect humans sporadically (as reported by the WHO: [https://www.who.int/publications/m/item/cumulative-number-of-confirmed-human-cases-for-avian-influenza-a\(h5n1\)-reported-to-who-2003-2024](https://www.who.int/publications/m/item/cumulative-number-of-confirmed-human-cases-for-avian-influenza-a(h5n1)-reported-to-who-2003-2024)). Therefore, even if prediction performance is high at the Orthomyxoviridae family level, it is possible for the model to fail in predicting the infectivity of H5Nx. We believe that this confusion also originated from the misleading data number for each category in Supplementary Figure 5, so we have clarified the number of virus data points (please note that approximately 5,500 sequences of H5Nx influenza viruses have been collected from animals after 2018, whereas only 52 sequences have been collected from humans).

Line 358 - 360: What are the valuable perspectives on enhancing generalization capabilities that were identified? Without identifying the causes of a lack of generalization (see comments for line 243 - 246), how can we resolve them?

Response: Following the reviewer's comment, we realized that the previous expression overemphasized the implications of our study, so revised it as follows:

Line 364-366: Our study provides a comprehensive benchmark for viral infectivity prediction models and highlights unresolved issues in fully exploiting machine learning to prepare for future zoonotic threats.

Furthermore, we apologize for repeating ourselves, but we would like to once again explain this study's achievements and future work. While viral infectivity prediction models have been developed in previous studies, no paper has pointed out the existence of difficult-to-predict viral lineages. We believe that highlighting these challenges in model performance provides valuable insights for improving models and preparing for zoonotic viruses.

Additionally, we also attempted to investigate the causes behind such prediction difficulties, but we could not identify clear reason at this moment. We agree that classical viral evolutionary analysis can help explain these prediction difficulties, as pointed out by the reviewer, and we have mentioned such future research directions in the manuscript as follows:

Line 334-341: We identified difficult-to-predict viral lineages, but we found no problems with these viral genera in the training datasets; the reason behind this difficulty of prediction thus remains unclear (**Supplementary Fig. 4d**). It may be that frequent changes in virus infectivity during Flavivirus evolution might have hindered model training (**Fig. 4**); however, the association between prediction difficulty and changes in infectivity should be further assessed through virus-host evolutionary analysis. Another possibility is that capturing infectivity changes due to a small number of genomic variations might be challenging for models using nucleotide sequences as the input.

Line 441 - 443: Please explain how and why this downsampling was done. As written, it sounds like poorly-predicted families are being removed, but what test data are being used to evaluate this performance? Are subsequent performance metrics still reliable when this is done?

Response: We apologize for the confusion, but we have not ruled out these viruses from our comparisons. Down-sampling is a dataset adjustment method used to address class imbalance, and it has widely used in machine learning model training. In this study, we performed down-sampling for viral families with PR-AUC < 0.75 in the initial dataset (Rhabdo, Circo, Pox, Hantaviridae) because the proportion of human-infecting viruses was low, except for Poxviridae.

Reviewer #2 (Remarks to the Author):

The authors present a thoroughly constructed study, particularly in its innovative use of large language models, and the systematic validation protocols.

Response: We are pleased that this reviewer appreciates our work and thank the reviewer for the constructive comments that have helped improve our manuscript.

Main issues:

Labelling of data:

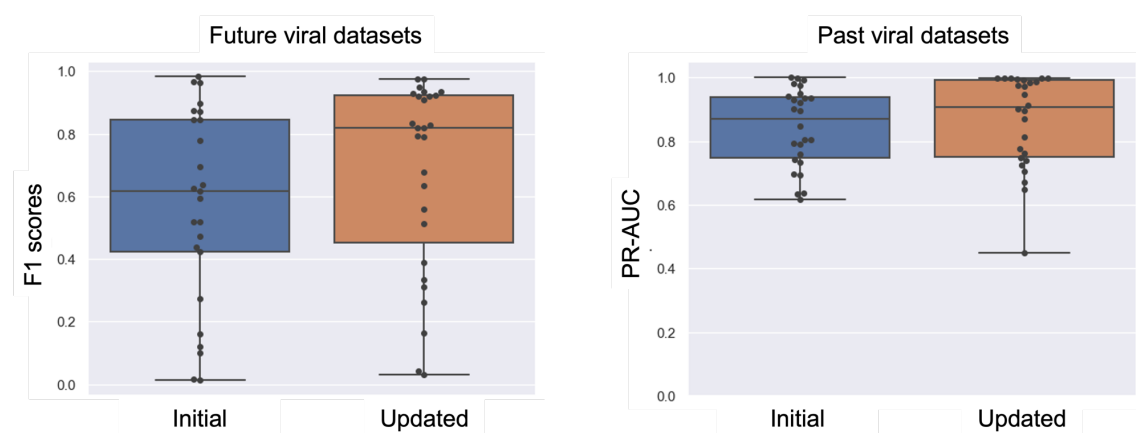
As the authors state: “Because the human infectivity of many viruses has not been directly validated, the infectivity label (i.e., human-infecting or animal-infecting) for each viral strain was defined based on the host information from which the viral sequences were collected”, have the authors verified (either all or subsample) their virus-host data to identify whether the associations are more likely to be virus infects/affects host, rather than virus associates with host. For instance, a virus may have had e.g. humans labelled as the host, but the sample could have been from stomach content? Have the authors accounted for this in their data?

Response: We appreciate the reviewer for pointing out our insufficient explanation. As per the reviewer’s comment, the NCBI Virus database contains not only sequences obtained from virus-infected samples but also virus sequences from environmental samples. To address this issue, we excluded viral sequences from environmental samples when downloading data from the NCBI Virus database. We have added this point to the Methods section as follows:

line 373-375: In addition, viral sequences collected from environmental samples were excluded from our dataset because the host organisms were ambiguous.

Generalisability: I am not sure if the hypothesis that “a queried virus would have the same infectivity as the most similar viral sequence” or if that hypothesis works particularly well with emerging viruses.

We appreciate this reviewer's constructive comment. Reviewer 1 also pointed out that the initial BLAST model may be a weak baseline model due to the "unpredictable" label (if a virus showed both human- and animal-infecting viruses with the same bit score in the BLASTn model, we judged it "unpredictable"). Therefore, we updated the BLAST baseline model to "predict human-infecting viruses if any human-infecting virus is present among the hit sequences." As seen below, this update improved the prediction performance of the BLAST baseline model for both past and future viral datasets. However, it is important to note that the updated BLAST baseline model did not outperform machine learning models, including the BERT-infect models.



We have added the following descriptions related to the BLAST baseline model:

Lines 157-159: BLASTn was used to assess the potential predictability of our datasets based on the simple hypothesis: if a human-infecting virus was included among the hit sequences, it was predicted to be human-infecting (see Methods).

Lines 433-438: We used BLASTn to evaluate potential dataset predictability based on the simple hypothesis that a virus showing sequence similarity to human infectious viruses is predicted to be human infectious. BLASTn prediction was conducted using the training dataset as a database and test dataset as a query. The following cases were judged as unpredictable: (i) there was no hit sequence with an E-value of $>1e-4$, or (ii) the aligned length of the top hit sequence did not cover $>50\%$ of the query sequence.

As the authors mention, generalizability is correlated with virus sequence similarity. However, I am not certain curating viral lineages associated with zoonoses would enhance generalizability. How do the authors envision this to work? Particularly given the fact that our knowledge is extremely biased: we are more likely to study any virus species/genus/family if it contains a human-infecting/zoonotic virus than if it does not. And this knowledge is often constructed post an emergence event.

Response: We appreciate this reviewer's constructive comment. First, as also noted by the other reviewers, we reconsidered use of the term "generalization" and deemed it inappropriate because our aim was to determine how well the infectivity of viruses that emerge after model construction using data obtained by 2018 can be predicted. Therefore, we have removed this term throughout the manuscript.

Additionally, we fully agree with the reviewer's comment regarding the model's applicability to unknown viruses. Since infectivity prediction models are trained on existing data, there is a potential dilemma in that the models may overlook the human infectivity of viruses that have not caused zoonotic infections. Model applicability for such unknown, human-infecting viruses can be verified by taxonomic hold-out validation, and improving this generalization capability is a future research goal machine learning models should aim for.

Furthermore, relying on a machine learning model to prepare for future zoonotic viral pandemics is not realistic, and we believe that other approaches, such as virus surveillance in animals and virological experiments, are still extremely important.

Minor points:

For labelling of animal-infecting or human-infecting, this is applied at the sequence level, I suggest the authors revise the manuscript to clarify, as the same strain of a virus could be both animal and human infecting.

Response: We appreciate the constructive comment. As also pointed out by the other reviewers, we believe that multi-labeling could improve model performance. On the other hand, the taxonomic hierarchy (species, subspecies, etc.) used for multi-labeling is a critical

hyperparameter, and this issue has not been investigated, as pointed out by Mollenze et al. in their review paper. Therefore, it is necessary to conduct preliminary investigations on how much sequence variation leads to infectivity changes and then set the appropriate hierarchy for building predictive models. We have added the following statement regarding such future work to the Discussion section:

Line 302-308: Furthermore, the multi-labeling method, which explicitly indicates viruses that infect both animals and humans, may improve predictive performance for zoonotic viruses. On the other hand, an important hyperparameter for the multi-labeling method has not been investigated: the taxonomic hierarchy to assign labels. Future work will need to include preliminary investigations into the degree of sequence variation that leads to infectivity changes to identify the appropriate taxonomic hierarchy for multi-labeling.

The authors should note that there exist data sources that have harvested taxonomically mapped virus-host (pathogen-host) associations from GenBank.

Response: We appreciate the reviewer's constructive comment but could not check the details due to the lack of specific data source references. To the best of our knowledge, the Virus-Host Database taxonomically shows virus-host associations; however, this database is limited to viruses registered in the RefSeq viral database, and it does not cover the scale of GenBank.

Reviewer #3 (Remarks to the Author):

General Comment

This study provides valuable improvements for the development of ML models for the prediction of virus human infectivity (i) by providing a comprehensive, curated dataset that appears to generally improve model performance across several types of architecture (ii) through their benchmarking work, comparing their models to at least three other ones, (iii) by highlighting important limitations for human infectivity prediction overall. At this stage I have several comments:

Response: We are pleased that this reviewer appreciates our work, and we are grateful for the reviewer's constructive comments that have helped us improve our manuscript.

Major Comments

Definition of host infectivity in the curated dataset

In the curated dataset used for model fine-tuning, each sequence is associated to a specific host, which may lead to important biases in the input data. For most virus species, sequencing data is strongly biased as sampling is uneven across geographic origins and host species and as some hosts are more easy to sample/access than others and as sequencing capacities may vary across locations. This will likely result in a training dataset that does not reflect the true ecology of the virus, in particular for viruses with a complex ecological cycle including arboviruses and may explain the results obtained for e.g. the Flavivirus genus. How do the authors balance the representation of hosts for viruses with two (or more) host species?

Response: We appreciate the constructive comment. This comment relates to two points: (1) sampling bias in virus sequences and (2) methods of infectivity labeling.

Regarding the first point, we fully agree with the comment. The impact of sampling bias in virus sequences on viral infectivity prediction has also been discussed in previous studies (<https://doi.org/10.1098/rstb.2020.0358>, or <https://doi.org/10.1371/journal.pbio.3001135>). Especially, since such prediction models are trained on existing data, there is a potential dilemma in that the models may overlook the human infectivity of viruses that have not caused zoonotic infections. In future work, model applicability for new viruses at the subgenus and species level

can be verified by taxonomic hold-out validation, and improving this generalization capability is a research goal machine learning models should aim for. Furthermore, relying on a machine learning model to prepare for future zoonotic viral pandemics is not realistic, and we believe that other approaches, such as virus surveillance in animals and virological experiments, are still extremely important.

Regarding the second point, as also pointed out by the other reviewers, we believe the multi-labeling methods could improve model performance, which may be particularly effective in explicitly indicating the infectivity of arboviruses, such as Flavivirus. On the other hand, the taxonomic hierarchy (species, subspecies, etc.) used for multi-labeling is a critical hyperparameter, and this issue has not been investigated, as pointed out by Mollenze et al. in their review paper. Therefore, it is necessary to conduct preliminary investigations on how much sequence variation leads to infectivity changes and then set the appropriate hierarchy for building predictive models. We have added the following statement regarding such future work to the Discussion section.

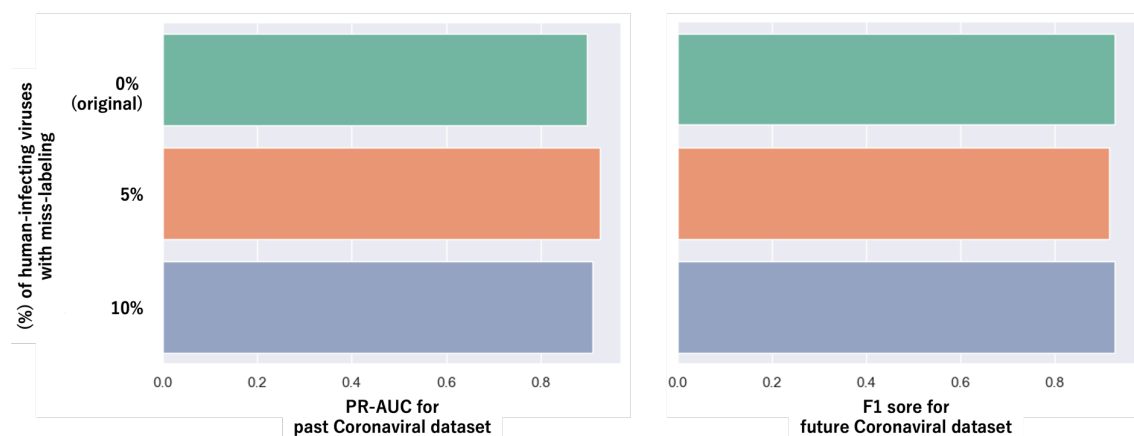
Line 302-308: Furthermore, the multi-labeling method, which explicitly indicates viruses that infect both animals and humans, may improve predictive performance for zoonotic viruses. On the other hand, an important hyperparameter for the multi-labeling method has not been investigated: the taxonomic hierarchy to assign labels. Future work will need to include preliminary investigations into the degree of sequence variation that leads to infectivity changes to identify the appropriate taxonomic hierarchy for multi-labeling.

Unidentified human-infecting viruses and false positives

The authors mention that they do not use experimentally validated data which results in two major limitations (i) input data labeled as not human for viruses with undemonstrated human-infection capacity ; (ii) a difficulty to discriminate between false and real positives. To better evaluate the validity of the authors' strategy (of using data not validated experimentally), could the authors try to evaluate the impact of having 5 or 10% additional unlabelled human-infecting viruses by artificially removing those labels for some known viruses ?

Response: We agree with the reviewer's concerns regarding the ambiguity of negative labels (i.e., viruses without human-infecting reports). Thus far, research on label noise has shown that model performance remains robust against incorrect labels (e.g., <https://doi.org/10.48550/arXiv.1705.10694>). Moreover, we found no significant changes in the predictive performance of the BERT-infect_{DNABERT} model for the family Coronaviridae, even for training on datasets in which 0% (i.e., the original result), 5%, or 10% of the human-infecting viruses were mis-labeled.

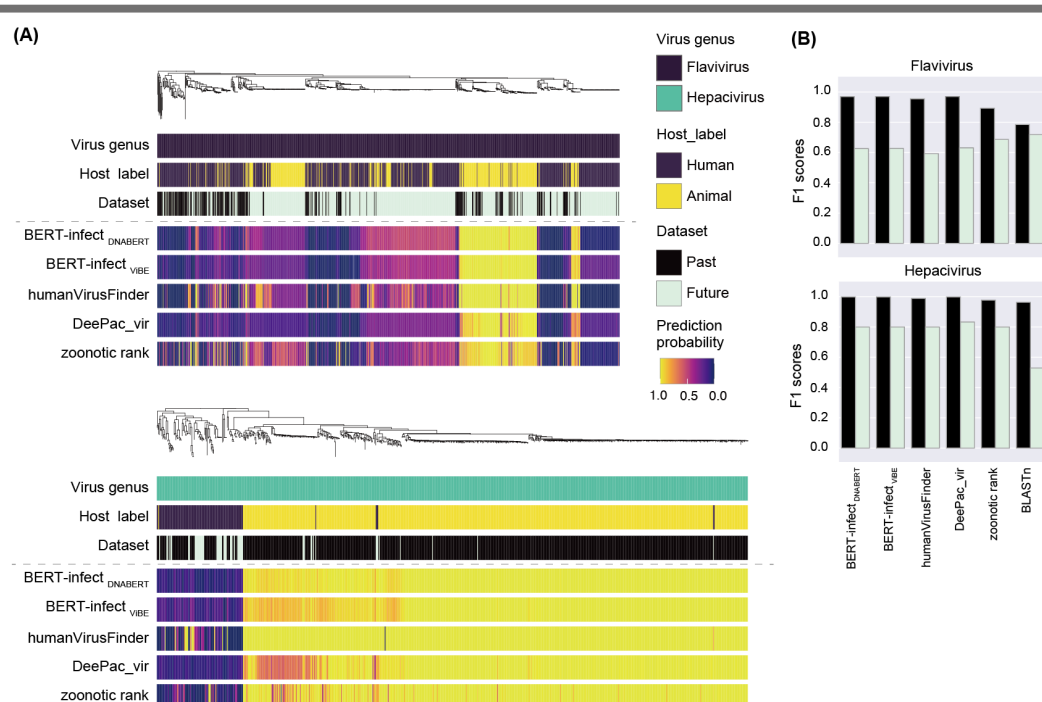
On the other hand, a reason for why the infectivity prediction model was robust to noisy data may be that our dataset contains similar viruses with the same infectivity (this is related to the limitation mentioned in the Discussion in **lines 309-319**). Thus, we believe that further verification is needed regarding the dataset construction methods.



Use of detailed viral phylogenies

The authors use viral phylogenies to compare F1 scores and host labels for the Flaviviridae and Coronaviridae families but the trees are difficult to read. For a genus as complex in terms of ecology as the Flavivirus genus, it would be beneficial to provide a more detailed view of the phylogeny to better identify the F1 score distribution along the tree.

Response: We appreciate the constructive comment. We have updated Fig. 4A and included genus-level trees.



Minor comments

Use of VIDHOP

It is unclear to me why the authors drop the VIDHOP model from their analysis.

Response: We apologize for any confusion. We excluded the VIDHOP model from this study because 1) a previous study, it was trained on a more than ten-fold larger dataset than that used in our study, and 2) it was constructed for only three viral families, making it an inappropriate comparison target for this research, which covers 26 viral families. The following revisions have been performed in the manuscript for clarity, including clarification for the VIDHOP model:

Lines 416-422: Table 1 lists the candidate models that served as comparisons in this study. Among these, the HostNet¹⁰ and VirusBERTHP¹¹ models were not publicly available, and retraining on our dataset was not possible. Furthermore, VIDHOP⁹ was excluded from the subsequent analysis for the following reasons: (i) the original study used >10,000 sequences per viral family as training dataset (i.e., approximately ten-fold the amount of our dataset), and (ii) the original VIDHOP model was only built for three viral families, making it difficult to compare model performance across a diverse range of viral families.

228-229 “These results demonstrate a high level of model generalization toward novel viruses”
Please, provide references to support your evaluation that the level of model generalization is indeed high.

Response: We apologize for the confusing statement. As shown in Fig. 3, we mentioned that almost all compared models, excluding DeePac_vir, showed equally high prediction performance in future viral datasets. These verifications were not conducted in previous studies, so no citations can be presented.

333-335 “Models using nucleotide sequences as inputs may effectively detect host adaptations in viral genome compositions; however, capturing infectivity changes due to minor genomic variations remains challenging.” The term "minor" is extremely context-specific, a single change can have a dramatic effect on a protein structure and the capacity to replicate within a given host. I recommend the authors reformulate towards a more objective term e.g. “a small number of genetic variations”.

Response: We thank the reviewer for the comment. We have modified this sentence as follows:

Line 339-341: Another possibility is that capturing infectivity changes due to a small number of genomic variations might be challenging for models using nucleotide sequences as the input.

Reviewer Comments:

Reviewer #1 (Remarks to the Author):

The authors have addressed my previous concerns, and the manuscript is much improved. I only have relatively minor comments remaining:

The new version introduces the phrase “newly emerged viruses” when referring to the future virus dataset. I think this over-states what this dataset represents, and by extension what the models evaluated are able to predict. All we know is that the specific genotypes are newly discovered / recently sequenced, there is no indication that these genotypes represent emerging viruses. For example, if someone sequenced and published a new rabies virus genome in 2020, then it would end up in the future virus test set. However, we already know about this virus more broadly (and it is probably represented in the training data), so the new genome does not make it a new virus, and certainly does not qualify it as an emerging virus.

Response: We are pleased that this reviewer appreciates our work and thank the reviewer for the constructive comments that have helped improve our manuscript.

We agree with the reviewer’s comment that the future viral datasets consisted of "viruses identified after model construction." Therefore, we have corrected the following revisions:

Line 49-50: Furthermore, models trained on data up to 2018 demonstrated robust predictive capability for most viruses identified post-2018.

Line 63-65: Our models trained on data up to 2018 demonstrated high predictive capabilities for most viruses identified after model construction.

Line 98-101: Third, given the successive emergence of novel zoonotic viruses, models should preferably be predictive for the infectivity of novel viruses (i.e., identified after the model construction)

Line 137-138: To assess the predictive capability against new viral sequences identified after the model construction,

Line 221-222: These results demonstrate a high level of model performance toward novel viruses identified after the model training.

Line 290-291: While models trained on the past virus datasets demonstrated high predictive capabilities for most viruses that identified after model construction,

Line 398-400: We divided the collected data into a past virus dataset for model training and a future virus dataset for evaluating model performance for novel viruses identified after the model construction (**Fig. 1B**).

The finding that current models do not predict zoonotic H5Nx influenza viruses (Discussion, line 324 – 325) should be mentioned in the results too, as it supports the point about SARS-CoV-2 also being missed by current models.

Response: Thank you for your insightful comments. We have moved the description of H5 influenza A viruses to the Results section and made the following changes.

Line 257: Challenges in detecting human infectious risk of emerging zoonotic viral lineages

Line 269-276: Furthermore, we also revealed that current models did not predict the human infectivity of most H5 influenza A viruses, which have reported >900 human infections so far, with an increase in spillover cases into dairy cattle, cats, and humans since March 2024 (Supplementary Fig. 5). Despite the high performance of several models, including the BERT-infect models, on past and future viral datasets, they failed to detect the human infection risk of H5 influenza A viruses. These findings emphasized an essential area for model improvement to accurately assess the risk posed by emerging zoonotic viruses.

Furthermore, we have removed redundant descriptions from the Discussion section.

Line 331-334: In this study, we also conducted large-scale predictions on the infectivity of zoonotic H5 influenza A viruses, showing that current models did not predict almost all their infectivity (Supplementary Fig. 5).

Other comments:

Line 129 – 132: Should this be <20 ? Otherwise the wording here is unclear.

Line 313 – 316. I think this cross-reference should be to supplementary figure S4.

Line 437: I think this should read $<1e-4$ (not $>1e-4$).

Line 532 – 533: Typo in “real-world”.

Response: Thank you for the reviewer’s careful checks. We corrected all the above descriptions.

Supplementary figure 1: From the caption, it is unclear whether points represent individual viruses (i.e., one per virus family), or a summary of predictions across multiple viruses in each family.

Response: We appreciate the reviewer for pointing out our insufficient explanation. We mentioned, “Each dot corresponds to a viral family.” in the caption of Fig S1. Additionally, please note that calculating the evaluation metrics in Fig. S1 requires multiple sequences.

Reviewer #2 (Remarks to the Author):

The authors have thoroughly responded to most of my initial comments, and I believe the manuscript is much better overall after the first round of reviews.

The only point that is still a bit ambiguous to me is in relation to labelling of data. Perhaps I was not so clear in my previous attempt, but I did not only mean environmental samples (those that are either without hosts, or labelled as environmental samples in GenBank), but also extends to samples with a well-defined, non-ambiguous host, e.g. in GenBank there exists few instances of plant viruses, and other non-human infecting viruses, that are associated with humans (e.g. host species indicated as human) due to them being sequenced from stomach content, or skin samples, etc. Have the authors considered this issue (in addition to removing environmental samples?). In such cases, the host will not be ambiguous, but the association with humans (or the host species) is simply commensal.

Response: We appreciate the reviewer's positive impression of our manuscript and the insightful and meaningful comments, which have helped us substantially improve the manuscript.

In this study, we used 26 viral families for the model construction and evaluation, all of which include human pathogenic viruses (described in lines 121-123). In other words, our datasets did not contain the viral families, only containing non-human infecting viruses that the reviewer is concerned about.

We recognize that the NCBI Virus database contains sequences obtained from humans for viral families such as Genomoviridae and Smacoviridae; however, we excluded these viral families due to the ambiguity of human pathogenicity.

Reviewer #3 (Remarks to the Author):

Main comment

The authors addressed the main concerns raised by the reviewers and revised their paper substantially, resulting in a paper that is now easier and clearer to read, with a better delineation of the achievements and perspectives of the study.

Specifically, the authors revised their paper by removing their claims regarding the generalisation capacity of their model in particular discussing its limits regarding the evaluation of infectivity for novel viruses. They also more thoroughly discuss the limits of their labeling approach and the need for improving labeling more generally to better handle the question of dual/multi-host viruses. Also, they re-evaluated their baseline based on the BLAST model to make it more realistic, and better detail their methods regarding the handling of segmented viruses and the curation of Genbank data. I still have a few comments that would need to be addressed for the manuscript to be suitable for publication.

Specific comments:

As discussed in the rebuttal in the response to several of the reviewer's comments, one of the main aims of this study is the improvement of sensitivity (ability to detect virus with human infecting potential) rather than specificity. I think briefly clarifying this point in the text would be beneficial to better depict the aims of this study.

Response: We are pleased that this reviewer appreciates our work and thank the reviewer for the constructive comments that have helped improve our manuscript.

We appreciate the reviewer for pointing out our insufficient explanation. As per the reviewer's comment, we added the following description:

Line 309-311: Given such limitations, this study focused on model sensitivity: detecting human infection risk, and animal-derived viruses predicted to be human-infectious by multiple models should be prioritized for risk assessment in future research (Supplementary Fig. 4).

As mentioned in one of the authors' responses, combining infectivity prediction to other approaches including surveillance in animals and experimentation will remain essential to preparedness against zoonotic threats. I suggest adding a few words on this question in the discussion to provide a broader perspective on the general integration of such approaches in a more tangible public health perspective.

Response: Thank you for your insightful comments. We added the following description to the Discussion.

Line 370-375: These findings emphasized an essential area for model improvement to accurately assess the risk posed by emerging zoonotic viruses. On the other hand, needless to say, relying on a machine learning model is not realistic to prepare for future zoonotic viral pandemics, and we believe that enhanced virus surveillance in animals and collaboration with virological experiments are still extremely important. The zoonotic risk surveillance should be enhanced through such a research ecosystem.

1320-322: "Furthermore, another limitation in our dataset curation is that the data labeling method cannot distinguish between infectivity (i.e., capability for animal-human infection) and transmissibility (i.e., ability to expand human-human infections)"

Of note, for animals feeding on other animals/humans, detection in a given organism may not even represent infection capacity (the virus can be detected in a blood meal of e.g. a mosquito that fed on an infected host but is resistant to infection), please could you reformulate the sentence accordingly ?

Response: Thank you for sharing an interesting model case regarding arbovirus infectivity. It should be noted that the primary goal of our models is to identify viruses with a potential risk of human infection, not to precisely determine infectivity in animals. This is also related to the

first comment made by this reviewer (i.e., model evaluation focusing on the sensitivity). Furthermore, since this case falls under the broader category of zoonotic viruses: the spread of infections to humans via animals, we believe that the human infection risk of such animal viruses should also be evaluated by our models.

Therefore, in lines 320–322 (now lines 329-331), we have kept the text focusing on human infectivity and human-human transmissibility. Additionally, we believe that the point raised by the reviewer is already covered in lines 302-306.